

GLOBAL THIRD-PARTY INSPECTION (TPI) - PIPE INSPECTION & TEST PLAN

Reference Standard: API Spec 5L / ASME B31.3 / ASME Section IX / ISO 10893 | Scope: Carbon & Alloy Steel Pipe

No.	Pipe Inspection and Test Plan Task	Vendor	TPI Inspection	Client	Reference Standard / Criteria
1	Mill test reports (incl. heat & product chemical analysis, tensile, CVN, heat treatment records)	H	R	R	API 5L Section 9.2 & 9.3 / EN 10204 3.1/3.2
2	WPS / PQR, Welders Qualifications (WQT records, competency certificates)	H	R	R	ASME Section IX / API 5L Annex D
3	Non-Destructive Testing (AUT / RT / UT / MT / PT) - Review of all examination reports and radiographs	H	W	R	ISO 10893 / ASME Section V / API 5L 9.12
4	Heat Treatment Execution & Verification (Normalizing, Quench & Temper, if specified)	H	W	R	API 5L Section 8.8 / Project Specification
5	Metallographic Examination & Macro/Micro Structure, if applicable	H	W	R	ASTM E3 / ASTM E112 / API 5L Annex H/J
6	Visual and Dimensional Control (OD, WT, Ovality, Straightness, Length, End Squareness, Bevel)	H	H	R	API 5L Section 9.11 & Table 10/11
7	Hydrostatic Pressure Test (%100 pipes, pressure recording verification)	H	H	R	API 5L Section 9.4 & 10.2.6 (100% SMYS / Holding time)
8	Surface Preparation, External 3LPE/FBE Coating, Marking, Stenciling and Color Coding	H	W	R	API 5L Section 11 / DIN 30670 / ISO 21809-1
9	Final Visual Inspection, Pipe End Protectors, Bundling, Packaging and Quantity Control	H	H	R	API 5L Section 14 / Purchase Order Specification
10	Final Documentation Review & Manufacturing Data Book Sign-off (Inspection Release Note - IRN)	H	H	R	EN 10204 3.1/3.2 / API 5L Section 13
11	Pre-Shipment Inspection & Loading Verification	H	H	R	API 5L Section 14 / Project Logistics Procedure

Legend: H: Hold Point (Mandatory witness/approval before proceeding) | W: Witness Point (Notification required, work may proceed if waived) | R: Document Review.

Inspection Severity: Class A = Full Resident Inspection | Class B = Stage Witness (Hydro, NDT, Final) | Class C = Random Witness & Doc Review | Class D = Final Review.